

# What is Acne?

Acne is a common skin condition that affects most people at some point. It usually starts during puberty but can develop for the first time in people in their late twenties and beyond (3% of adults have acne over the age of 35), and occasionally in younger children.

Acne most commonly develops on the face, back and chest ranging from several spots to a more severe problem causing scarring.



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## What Does Acne Look Like?

Acne causes oily skin, comedones (blackheads and whiteheads), pus-filled spots (pustules), larger deeper bumps (nodules), and discolouration that can be red or hyperpigmented (darker than your normal skin colour). Severe acne can also cause scars, which can be firm and raised (hypertrophic or keloid) or indented (pitted or atrophic).

## What Causes Acne?

The natural oils in our **skin (sebum)**, arise in **sebaceous glands**. The sebum then passes into hair follicles and flows out onto the skin surface.

In people with acne the sebaceous glands are oversensitive, even to normal levels of hormones, causing **too much sebum** to be produced along with **thickening** of the inner lining of the hair follicle leading to a **plugging effect**. If

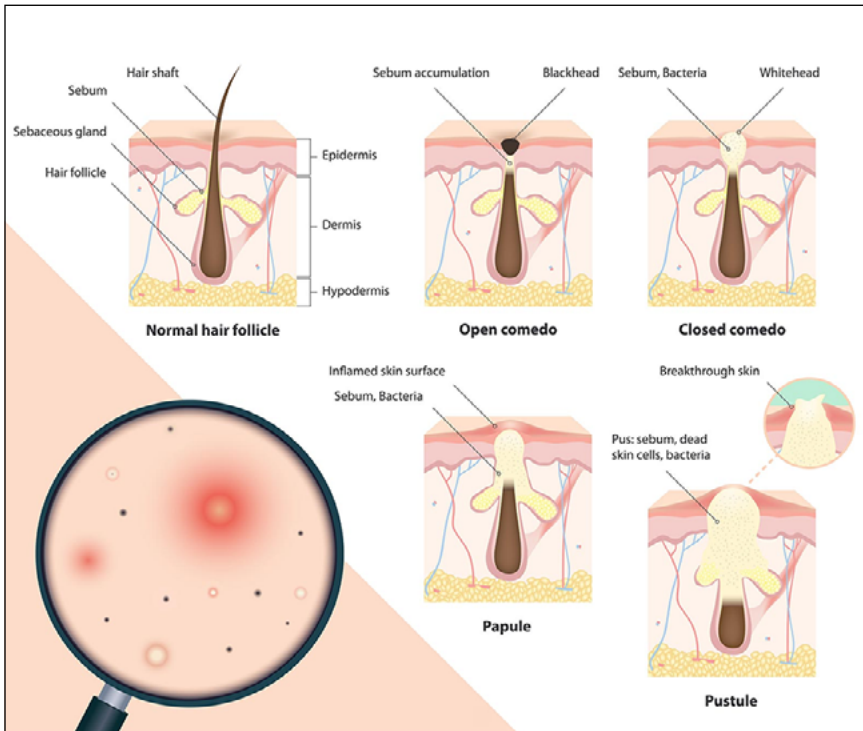
the plugged follicle is close to the surface of the skin, it bulges outwards, creating a whitehead. Alternatively, the plugged follicle can be open to the skin, creating a blackhead.

The changes described alter the activity of a usually harmless **skin bacterium called C. acnes (Cutibacterium acnes)**, which becomes more aggressive causing inflammation and pus. Despite of this, acne is NOT infectious, and cannot be passed on to other people.

## What Else Can Contribute to Acne?

- If your **parents, brothers/sisters** had moderate-severe acne, you are more likely to have troublesome acne
- **Hormones** – the vast majority of people with acne have **normal hormone levels**, however, hormones can impact certain patients:

- o **Menstrual periods** – some women have a flare-up of acne just before their period
- o **Pregnancy** – many women have symptoms of acne, usually during the first 3 months
- o **Some contraceptives** – the progestogen-only pill or contraceptive implant can sometimes make acne worse
- o **Polycystic Ovarian Syndrome** – a common condition in women that can cause acne, irregular periods, weight gain, excess hair growth in unusual sites (hirsutism) and/or hair thinning (alopecia). Consult your GP if you have such symptoms as further tests may be needed
- o Some hormone tablets taken illegally **eg anabolic steroids**
- Some other **medications** eg steroids, lithium and certain medications used to



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treat epilepsy – ask your GP if you think you may be on one of these medications

be squeezed or scratched as this may damage (scar) the skin

- **Diet** – food with a high glycaemic index (eg sugary foods, white bread, potatoes, white rice etc) have been shown to cause or aggravate acne in some people
- Some **cosmetic products** eg pomade hair products, which have a base of mineral oil or petroleum jelly and are more occlusive to the skin
- **Smoking** can contribute to acne in older people

### Things You Can Try If You Have Acne

- Wash the affected area with a mild soap or cleanser and lukewarm water, not more than twice a day
- While spots with a pus-filled head (pustules) can be gently squeezed to express the pus, other spots must not

- Avoid make-up, skincare and sun care products that are oil-based, instead use water-based products marked as 'non-comedogenic', as they're less likely to block the pores in your skin. Completely remove make-up before going to bed
- Eat more foods with a lower glycaemic index (fruit and vegetables, brown bread/rice/pasta)
- Sunbathing, sunbeds and sunlamps should not be used to manage acne for any length of time – they will not cure your acne, but they will age your skin (meaning more wrinkles) and increase your risk of skin cancer
- Stop smoking

- See your pharmacist and ask about over-the-counter acne products

### Make An Appointment At Your General Practice

- Treatment options provided by your GP include topical treatments (applied directly to the skin), oral antibiotics (taken by mouth) and oral contraceptive pills – for more information refer to <https://www.pcds.org.uk/desktop-treatment-guide#9-acne-vulgaris>

- Notes on these treatments:

- o **Topical treatments** should **always be part of your treatment plan** as they help unblock the plugged pores. These treatments can cause significant dryness and irritation and so should be **used initially by applying to the skin and washing off after 30 minutes**. Over the course of a few weeks you should increase how long the treatment is left on, the length of time that any person can tolerate varies significantly eg some people are only able to manage 1-2 hours, whereas others can put on in an evening and wash off in the morning. If the skin is dry, you can use **an oil-free moisturiser through the day**. If you are using any product containing **benzoyl peroxide** it can bleach pillowcases and clothing if not careful (but not the skin)

- **Oral antibiotics** – tend to be used for **3-6 months** at a time and must always be **combined with a topical treatment** that does NOT contain an antibiotic. Some antibiotics should not be taken at the same time as food or can make your skin more sensitive to the sun, so read the instructions carefully. It is also essential to let your doctor know if you are **planning a pregnancy** as some antibiotics cannot be taken if you are pregnant. There is no current evidence to suggest that long course of the type

of antibiotics used in acne are harmful, however, they will reduce the number of normal bacteria in your gut, especially in the first few weeks of treatment. You may wish to consider taking Probiotics products; live bacteria and yeasts that are usually added to yoghurts or other food products, although there is no current evidence that this is beneficial (although no evidence doesn't mean that they are not helpful – it means we don't know)

- **Hormonal treatments** – some contraceptives are more 'acne friendly' than others, which can be discussed with your GP. Occasionally other hormone blocking treatments such as spironolactone are used. For more information refer to <https://www.pcds.org.uk/desktop-treatment-guide#9-acne-vulgaris>

### Referral To A Specialist

The following people may need to be referred to a specialist in acne, who will be either a Dermatologist or a GPwER in Dermatology (a GP who has also been trained in some aspects of dermatology such as acne):

- Patients with severe acne eg scarring acne – should be started on treatment and referred urgently
- Patients with long-term acne that it not responding adequately to at least two different oral antibiotics in combination with topical treatments
- Patients with marked hyperpigmentation (skin colour darker than their natural skin colour)
- Significant psychological upset (see below)

### Acne And Mental Health

Acne has a negative psychological impact and can affect your mental health. If your acne is not improving with treatment and is having a major impact on your mental health, then you should ask your GP to be referred. If you ever feel **very depressed or suicidal about your skin then you must ask for urgent help – speak to family members, phone your GP or the Samaritans (116 123 from any phone) and remember there is ALWAYS HOPE; MUCH CAN BE DONE TO HELP YOU AND YOUR SKIN.**

### What Will Happen When You Are Referred?

- You will be referred for consideration of a drug called **isotretinoin**. Isotretinoin is an **excellent treatment for moderate-severe acne**. In some patients only one course is required, in others the acne may come back at some future point requiring a further course of treatment. One of the reasons why a specialist prescribes isotretinoin is that it has been associated with **some serious side effects**, although it must be remembered that these are **extremely uncommon**, however, it is important you **understand all the risks and benefits of treatment**. The following link provides more information <https://www.bad.org.uk/shared/get-file.ashx?id=2314&item type=document>
- Your GP will also need to organise **some bloods tests** (FBC, U+E, LFT, lipid profile)
- **Female fertile patients also need to be taking a suitable contraceptive agent** as isotretinoin is damaging to an unborn child (teratogenic). If there are exceptional circumstances meaning

that you are not on contraception your GP will need to state this on the referral letter. It is important to remember that **isotretinoin does NOT affect** your chances of having children in the future

### Scarring And Hyperpigmentation (Areas Of Dark Skin Colour)

Most patients will not end up with **scarring**. For those that do there is no medicine (tablet) that can help treat scarring. Once your acne treatment has finished your will get some natural improvement in the appearance of your skin for 6-12 months afterwards. If you are still unhappy about the appearance of your skin after this time, then some patients will benefit from physical treatments such as laser, or steroid injections into thick scars – in such cases ask for a referral to a dermatologist to see if any treatment may help. Beware that treatments such as laser are **generally unavailable on the NHS**.

Skin camouflage is an inexpensive and effective method for **disguising changes in the pigmentation of the skin** which can sometimes remain after acne has been treated – ask your GP to refer you to a camouflage clinic if there is one in your local hospital, otherwise you can contact <https://www.changingfaces.org.uk/>.